

Basic principles over modulators

- Introduction
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- Silicon Photonics Modulators
- Basic electronics concepts
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Introduction:

- We use silicon waveguides to carry light signals on chips. Optical modulators in these circuits encode data by changing light's phase or amplitude, we need modulators because lasers alone cannot carry information at high speed and with high fidelity — modulators encode electronic data onto light for communication, sensing, and computation.

Introduction:

- In optical transmitters, modulation is generally performed either by directly modulating the laser, i.e., by turning on and off the laser, or by modulating the light out of the laser using an external modulator.
- **Direct modulation**

Switching the laser intensity by changing the driving current to the laser. Good for simple and cheap applications, but not suitable for high speed and long distance modulation due to switching effect of semiconductor technology, and the chirping effect (chirping effect is the instability in signal frequency due to variation of laser intensity).

Introduction:

- **External modulation**

External modulation is caused by supplying the light out of the laser source into modulators, and then into the channel, modulators alter the properties of the light signal.

Good for high speed and less distortion applications, but it's very complicated method, more expensive than direct modulation, multiple variables must be considered before using this method (like wavelength variation, temperature variation, industry fabrication imperfections, etc.).

Types of External Modulators:

- Electro-Optics Modulation (Amplitude – Phase - MZI)

Type of external modulator that uses electric field to modify the refractive index of waveguide, in which it changes the properties of the optical signal passing through the wave guide, the properties in effect are amplitude, phase, frequency, polarization or a combination of the previous elements.

Types of External Modulators:

- Acousto-Optic Modulation (Amplitude – Frequency – Phase _ Beam Deflection)

Type of external modulators that uses Bragg diffraction principle to control properties of light passing through certain material using acoustical waves, the properties that can be modified are amplitude, phase, frequency and beam deflection.

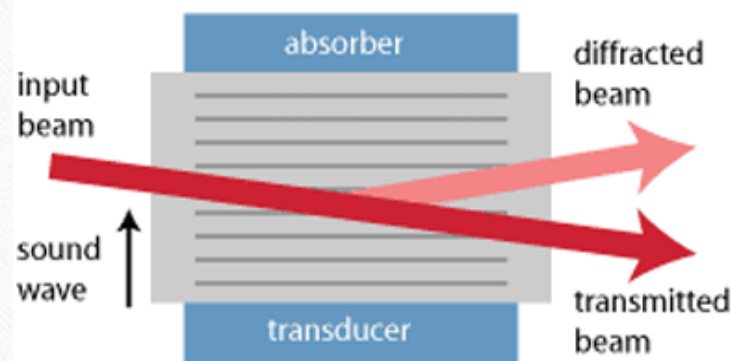


Fig.1

Types of External Modulators:

- Electro-Absorption Modulation (Amplitude – **Phase** - wavelength)

Type of external modulators that controls the amplitude and wavelength spectrum of the optical signal by controlling the absorption rate of the waveguide material using electrical field.

- Magneto-Optic Modulation (Amplitude - Polarization)

Type of external modulators that controls the amplitude and polarization using magnetic fields by the principle of faraday effect.

Types of External Modulators:

- Thermo-Optic Modulation (Phase – Amplitude - Wavelength)

Type of external modulators that controls refractive index of waveguide using heat energy.

Types of External Modulators:

Type	Modulation Method	Common Material	Application
Electro-Optic (EOM)	Refractive index (electric field)	• LiNbO₃ • GaAs	• High-speed telecom
Acousto-Optic (AOM)	Acoustic wave diffraction	• Quartz • TeO₂	• Laser scanning • Spectroscopy
Electro-Absorption	Light absorption	• InGaAs • InP	• Compact modulators in fiber optics
Magneto-Optic	Polarization (magnetic field)	• Garnets	• Optical isolators • sensors
Thermo-Optic	Heat-induced index change	• Silicon • Polymers	• Integrated photonics

Silicon Photonics Modulators

- Mach-Zehnder Interferometer Modulators
- Micro Ring Resonator Modulators
- Electro-Absorption Modulators
- Carrier Injection/Depletion based Modulators
- Thermo-Optic Modulators

Silicon Photonics Modulators

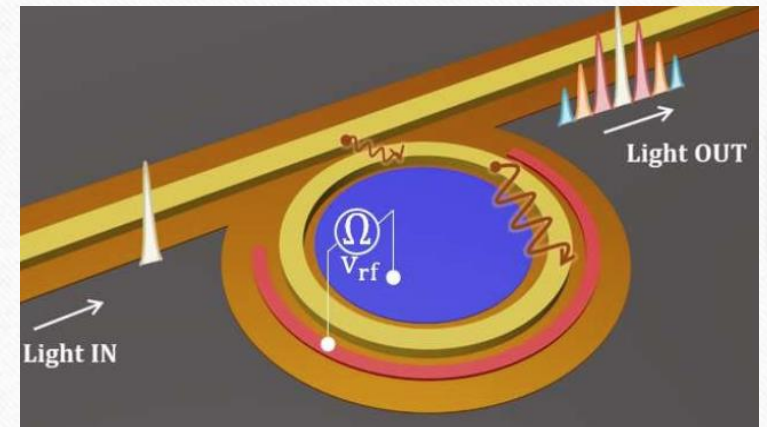
- Micro Ring Resonator Modulators

Low insertion loss / energy efficient /

low size thus they suit for high density fabrications.

Fabrication sensitive /

temperature variation sensitive thus they require heaters and controlling circuits.



Silicon Photonics Modulators

- Mach-Zehnder Interferometer Modulators

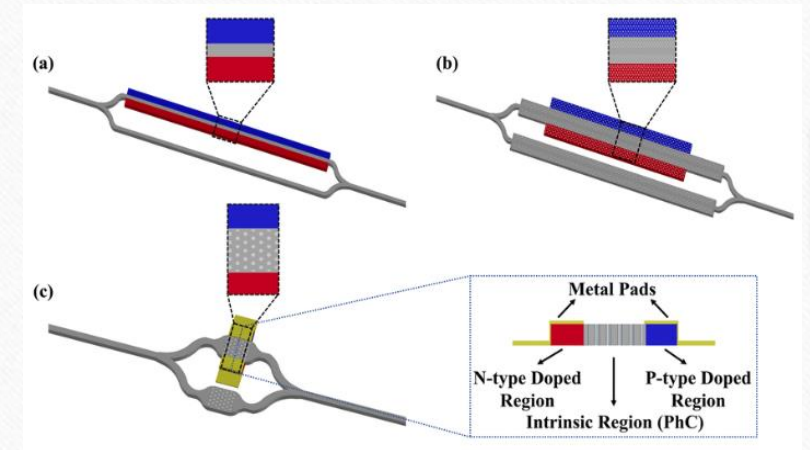
Phase shift equation is described by:

$$\Delta\phi = [\beta]L = \left[\frac{2\pi}{\lambda_0}\Delta n_{eff}\right]L$$

Can sustain temperature changes up to 8.3 C.

Large in size compared to the micro ring modulator.

Types: 2X2 1X2 1X1 / symmetric asymmetric /
balanced unbalanced



Silicon Photonics Modulators

- Carrier injection/depletion modulators

The impurity and carrier distributions in a carrier-depletion phase modulator are illustrated in Figure:

p–n Junction in a rib waveguide: (a) cross-sectional view of the impurity distribution assumed in the abrupt junction model;

(b) cross-sectional view of the carrier distribution;

(c) 1D free carrier distributions.

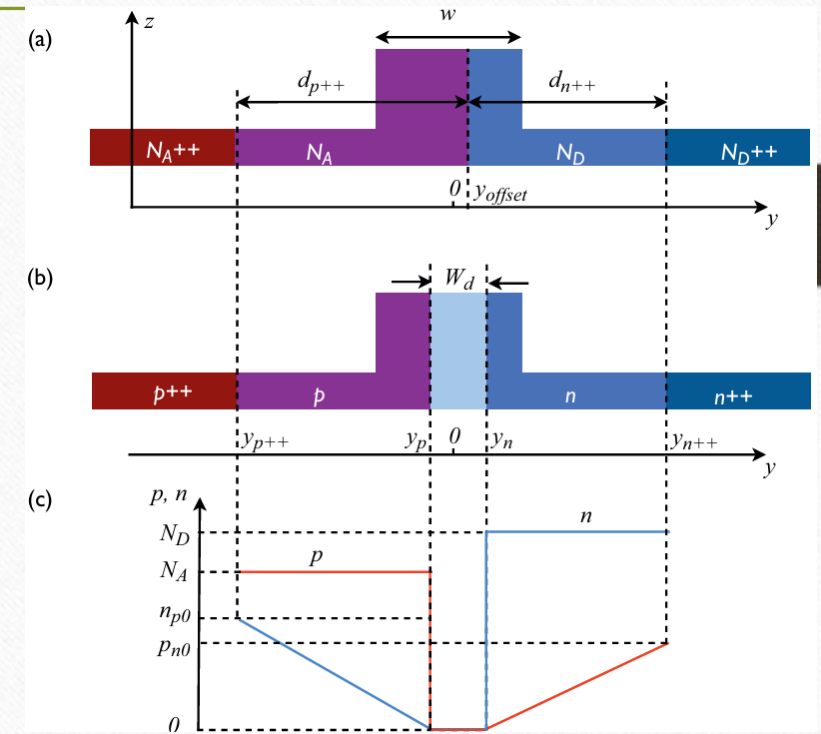


Fig.2

Silicon Photonics Modulators

- The width of the depletion region, W_d , is determined by the impurity densities (N_A and N_D), as well as the applied voltage (V), and is given by:

$$W_d = \sqrt{\frac{2\epsilon_0\epsilon_s(N_A + N_D)(V_{bi} - V)}{qN_A N_D}}$$

where ϵ_s is the relative permittivity and V_{bi} is the built-in or diffusion potential of the junction given by:

$$V_{bi} = \frac{k_B T}{q} \ln \frac{N_A N_D}{n_i^2}$$

Electronics concepts relating modulators with silicon photonics

- Knowing that Silicon is a semiconductor, doping creates p-type (holes) and n-type (electrons) regions. A p–n junction diode in silicon can be forward- or reverse-biased to inject or remove free carriers.

Free electrons/holes in the waveguide change its refractive index (and absorption) via the plasma dispersion effect (where the concentration of carriers is varied either by injecting or removing carriers from the device. This effect will be used for high-speed p–n junction modulators) ^[1] .

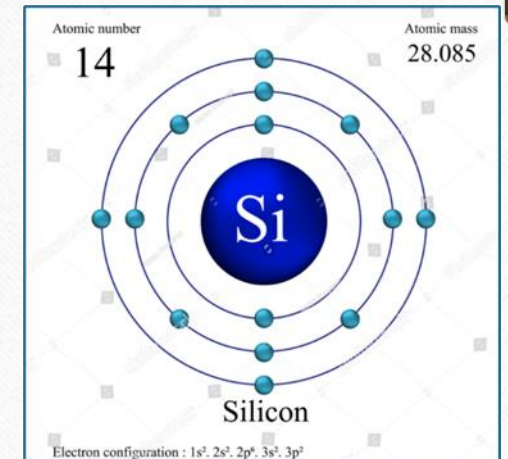


Fig.3

Electronics concepts relating modulators with silicon photonics

- A p-n junction diode

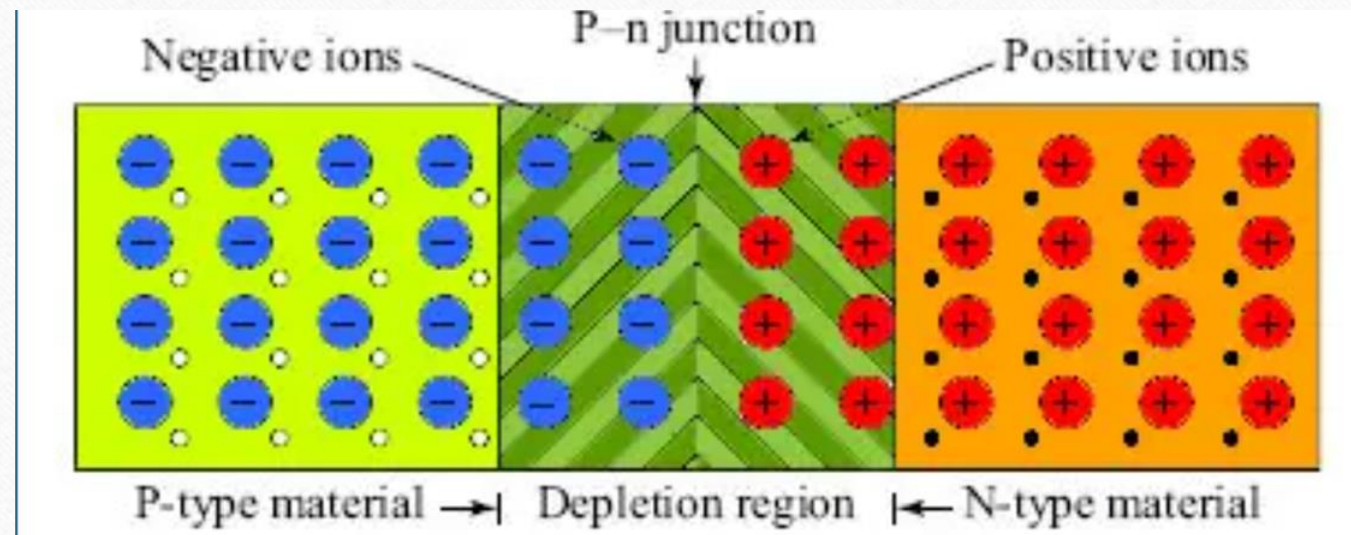


Fig.4

Electronics concepts relating modulators with silicon photonics

- The change in index of refraction is phenomenologically described by [3]:

$$\Delta n \text{ (at 1550 nm)} = -8.8 \times 10^{-22} \Delta N - 8.5 \times 10^{-18} \Delta P^{0.8}$$

$$\Delta n \text{ (at 1310 nm)} = -6.2 \times 10^{-22} \Delta N - 6 \times 10^{-18} \Delta P^{0.8},$$

and the change in absorption is described by:

$$\Delta \alpha \text{ (at 1550 nm)} = 8.5 \times 10^{-18} \Delta N + 6 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]}$$

$$\Delta \alpha \text{ (at 1310 nm)} = 6 \times 10^{-18} \Delta N + 4 \times 10^{-18} \Delta P \text{ [cm}^{-1}\text{]},$$

where N, P are the carrier densities of electrons and holes [cm⁻³].

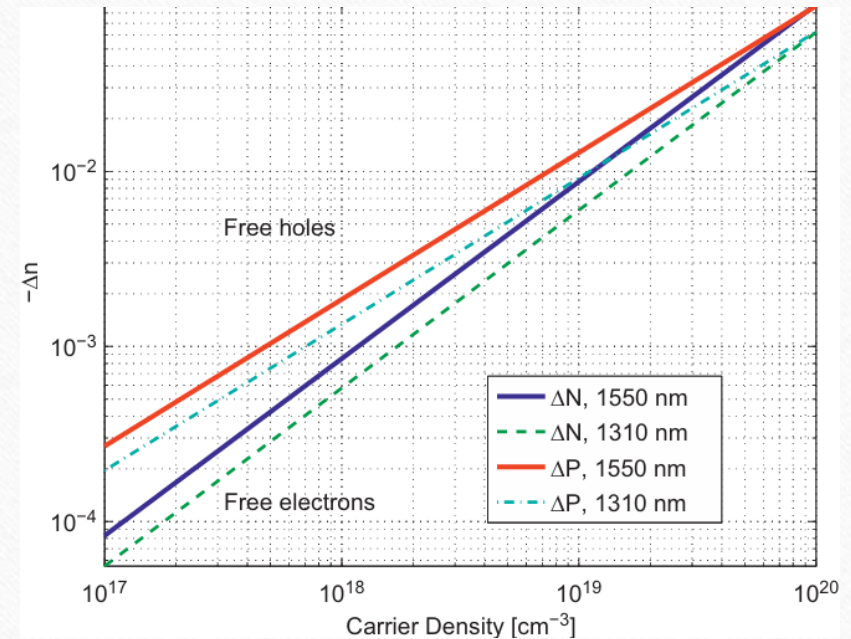


Fig.5

Electronics concepts relating modulators with silicon photonics

- Forward-biasing a p–n junction (or PIN diode) injects electrons and holes into the waveguide (increase ΔN) – this raises index and absorption. Reverse-biasing depletes carriers (reducing ΔN) – this lowers index (and reduces absorption) [4].
- Although the modulation is very weak below the diode threshold (also referred to as the cut-in or knee voltage). However, just a small voltage beyond threshold injects enough carriers to obtain a π phase shift [5] .

Electronics concepts relating modulators with silicon photonics

- Similar to the p–n junctions considered above, active silicon photonics devices can be constructed using forward bias PIN junctions. In this case, it is useful to increase the size of the junction by introducing an un-doped (intrinsic) region between the P and N regions, as shown in Figure:

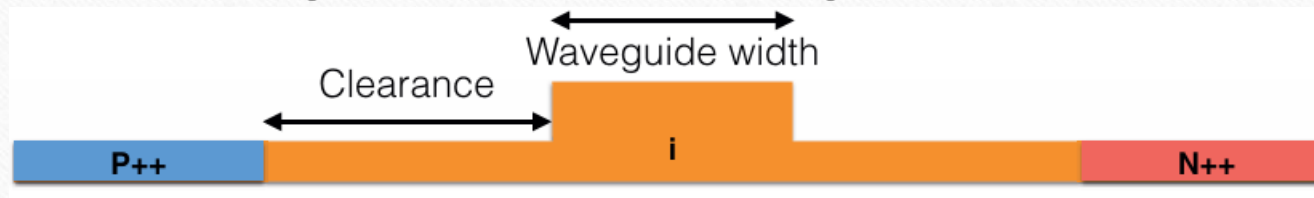


Fig.6

Electronics concepts relating modulators with silicon photonics

- The carrier concentration for the plasma dispersion effect in silicon can be modulated by various structures. The most commonly studied silicon optical modulator structures are based on carrier accumulation, carrier injection, or carrier depletion. Each structure must be able to guide light while allowing to modulate the carrier concentration in the region of the optical mode [5].

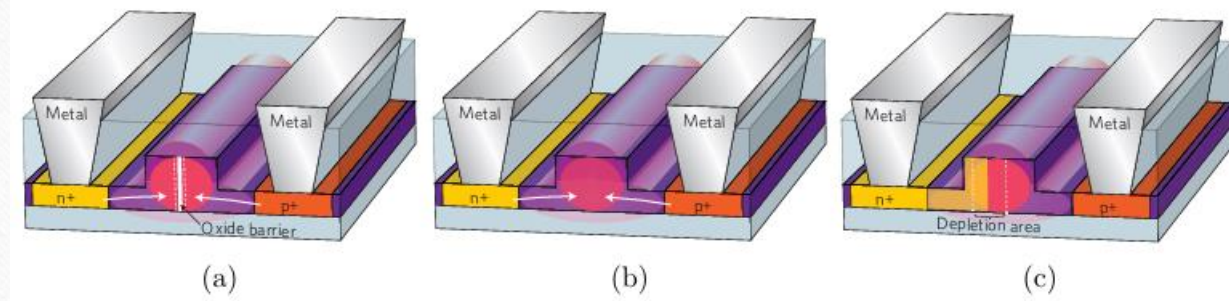


Fig.7

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Thanks
